

2(a)	a body continues at (rest or) constant velocity unless acted upon by a resultant force	B1
2(b)(i)	distance = $[\frac{1}{2} \times (2.0 + 4.4) \times 3.0] + [4.4 \times 2.0]$	C1
	= 9.6 + 8.8	A1
	= 18 m	
2(b)(ii)	$a = (v - u) / t$ or gradient or $\Delta v / (\Delta)t$	C1
	e.g. $a = (4.4 - 2.0) / 3.0 = 0.80 \text{ m s}^{-2}$	A1
2(b)(iii)	1. force = $240 \cos 28^\circ$ or $240 \sin 62^\circ$ = 210 N	A1
	2. resultant force = 89×0.80 (= 71.2 N)	C1
	$R = 210 - 71$ = 140 N	A1
2(b)(iv)	$T \sin 45^\circ = mg$	C1
	$T = (89 \times 9.81) / \sin 45^\circ$ = 1200 N	A1